

Kali Linux System Enumeration Report

Labs 1-6 — Section 3 Practical Exercises

1. Introduction

This report summarizes six hands-on labs performed in a Kali Linux virtual machine. The goal was to practice system enumeration: identifying the operating system, desktop environment, package repositories, running services, hardware, and firmware. Enumeration is the first step in both security testing and system hardening. All results below come from the screenshots taken during the labs.

2. Environment

Item	Value
Operating system	Kali GNU/Linux Rolling 2026.2 (Debian-based)
Kernel	6.19.14+kali-amd64, x86_64
Virtualization	VirtualBox
Desktop / session	XFCE, X11, LightDM
Resources	1 vCPU, about 1.9 GiB RAM, 19 GB disk

3. Lab 1 — Identify the System

Commands: `uname -a`, `uname -r`, `cat /etc/os-release`, `hostname`, `hostnamectl`

What I did: Checked the kernel, OS release, and hostname, then ran `hostnamectl` for platform details.

Result: The system is Kali Rolling 2026.2 with kernel 6.19.14, hostname `kali`, running as a VirtualBox virtual machine. The 2006 firmware date shown is VirtualBox's emulated BIOS, not real hardware.

```
—(kali@kali)-[~]
_$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

—(kali@kali)-[~]
_$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

—(kali@kali)-[~]
_$ uname -r
6.19.14+kali-amd64

—(kali@kali)-[~]
_$ hostname
kali

—(kali@kali)-[~]
_$ hostnamectl
Static hostname: kali
Icon name: computer-vm
Chassis: vm
Machine ID: [REDACTED]
Boot ID: [REDACTED]
Virtualization: oracle
Operating System: Kali GNU/Linux Rolling
Kernel: Linux 6.19.14+kali-amd64
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 10month 4d
```

Lab 1 evidence: OS, kernel, and hostnamectl output

4. Lab 2 — Desktop Environment

Commands: `echo $XDG_CURRENT_DESKTOP`, `echo $DESKTOP_SESSION`, `loginctl show-session "$XDG_SESSION_ID" -p Type`

Result: The desktop is XFCE, the session is `lightdm-xsession`, and the session type is X11.

Problems and fixes:

- `echo $DESKTOP_session` returned nothing because variable names are case-sensitive. Correct form: `$DESKTOP_SESSION`.
- `echo $XDG_DESKTOP_TYPE` also returned nothing because that variable does not exist. The session type was read with `loginctl` instead.

```
(kali@kali)-[~]
$ echo $XDG_CURRENT_DESKTOP
XFCE
(kali@kali)-[~]
$ echo $DESKTOP_SESSION
lightdm-xsession
(kali@kali)-[~]
$ echo $XDG_DESKTOP_TYPE
(kali@kali)-[~]
$ loginctl show-session "$XDG_SESSION_ID" -p Type
Type=x11
(kali@kali)-[~]
$ echo $XDG_CURRENT_DESKTOP
XFCE
(kali@kali)-[~]
$ echo $DESKTOP_SESSION
lightdm-xsession
(kali@kali)-[~]
$ echo $XDG_CURRENT_DESKTOP
XFCE
(kali@kali)-[~]
$ echo $DESKTOP_SESSION
lightdm-xsession
```

Lab 2 evidence: desktop and session checks

5. Lab 3 — APT Repositories

Commands: `cat /etc/apt/sources.list`, `ls -la /etc/apt/`, `ls -la /etc/apt/sources.list.d/`, `grep -R "^deb" /etc/apt/2>/dev/null`, `cat /etc/apt/sources.list.d/kali.sources`

Result: There is no `/etc/apt/sources.list` on this system. Repositories are defined in one file, `/etc/apt/sources.list.d/kali.sources`, which uses the newer `.sources` format:

Field	Value
Types	deb
URIs	http://http.kali.org/kali/
Suites	kali-rolling
Components	main contrib non-free non-free-firmware
Signed-By	/usr/share/keyrings/kali-archive-keyring.gpg

Problems and fixes:

- A typo (`source.list`) in a `grep` command produced no visible error because `2>/dev/null` hid it.
- `cat ../kalisources` failed with "No such file or directory" because the dot was missing. The correct name is `kali.sources`.
- `sources.list` was missing, which is normal here. No file was created.

Lesson: Hiding errors with `2>/dev/null` can mask mistakes. Check filenames with `ls` first.

```
(kali@kali)-[~]
$ cat /etc/apt/sources.list.d/kali.sources
# See https://www.kali.org/docs/general-use/kali-apt-sources/
Types: deb
URIs: http://http.kali.org/kali/
Suites: kali-rolling
Components: main contrib non-free non-free-firmware
Signed-By: /usr/share/keyrings/kali-archive-keyring.gpg
```

6. Lab 4 — Services and Network Exposure

Commands: `systemctl --type=service --state=running`, `systemctl status ssh`, `systemctl is-enabled ssh`, `sudo ss -tulpn`

Result:

- 20 services were running, typical of a desktop VM (NetworkManager, LightDM, D-Bus, cron, VirtualBox guest utilities, and others).
- SSH is installed but **inactive and disabled**.
- `ss -tulpn` showed **no listening TCP or UDP ports**.

Takeaway: A running service does not always listen on a network port. With nothing listening, this machine exposes no network services.

```
(kali@kali)~$ systemctl status ssh
o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)

(kali@kali)~$ systemctl is-enabled ssh
disabled

(kali@kali)~$ sudo ss -tulpn
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
(kali@kali)~$
```

Lab 4 evidence: SSH disabled, no listening ports

7. Lab 5 — Hardware Enumeration

Commands: `lspci`, `lsusb`, `lscpu`, `free -h`, `df -h`

Result:

- Devices are virtual: VMware SVGA adapter, Intel 82540EM network card, VirtualBox Guest Service, and a VirtualBox USB Tablet.
- CPU: 11th Gen Intel Core i7-1165G7, with 1 vCPU assigned.
- Memory: 1.9 GiB total, 1.1 GiB available, swap unused.
- Disk: root filesystem `/dev/sda1` is 85% full (15 GB of 19 GB).
- `lscpu` also listed CPU vulnerability status. Most entries were "Not affected" or mitigated; Spec store bypass was reported as "Vulnerable".

```
(kali@kali)-[~]
└─$ lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:02.0 VGA compatible controller: VMware SVGA II Adapter
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)
00:04.0 System peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Service
00:05.0 Multimedia audio controller: Intel Corporation 82801AA AC'97 Audio Controller (rev 01)
00:06.0 USB controller: Apple Inc. KeyLargo/Intrepid USB
00:07.0 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:08.0 USB controller: Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family) USB2 EHCI Controller
00:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA Controller [AHCI mode] (rev 02)

(kali@kali)-[~]
└─$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 002 Device 002: ID 80ee:0021 VirtualBox USB Tablet

(kali@kali)-[~]
└─$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 1
On-line CPU(s) list:    0
Vendor ID:              GenuineIntel
Model name:             11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz
CPU family:             6
Model:                 140
Thread(s) per core:     1
Core(s) per socket:     1
Socket(s):              1
Stepping:               1
Microcode version:      0xffffffff
BogoMIPS:               5606.40
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fx
sr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc c
puid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes x
save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ibrs_enhanced fsgsbase bmi1
avx2 bmi2 invpcid rdseed adx clflushopt sha_ni arat md_clear flush_l1d arch_capabilities

Virtualization features:
Hypervisor vendor:      KVM
Virtualization type:    full
Caches (sum of all):
L1d:                    48 KiB (1 instance)
```

Lab 5 evidence: `lspci`, `lsusb`, `lscpu`

8. Lab 6 — Firmware

Commands: `ls /lib/firmware`, `find /lib/firmware/ -type f | head -20`, `dmesg | grep -i firmware`, `sudo dmesg | head -30`

Result:

- `/lib/firmware` exists and holds many firmware files (Intel Wi-Fi, audio, and others).
- `dmesg` boot lines showed the kernel version, a BIOS memory map, NX protection active, and the VirtualBox hypervisor.
- A hibernate service was skipped because EFI variables were missing, which suggests the VM boots in BIOS mode (my inference from the output).

Problem and fix: `ls /lib/frimware` failed with "No such file or directory" because of a spelling mistake. Retyping it as `/lib/firmware` worked. The directory was never missing.

```
(kali@kali)-[~]
└─$ ls /lib/firmware
3com                               iwlfwifi-gl-c0-fm-c0-98.ucode     TAS2XXX103C8DE91.bin
8E86-1-9.bin                       iwlfwifi-gl-c0-fm-c0-c101.ucode   TAS2XXX10A40.bin
8E86-1-A.bin                       iwlfwifi-gl-c0-fm-c0-c102.ucode   TAS2XXX10A41.bin
8E86-1-C.bin                       iwlfwifi-gl-c0-fm-c0.pnvm         TAS2XXX11540.bin
8E86-1-D.bin                       iwlfwifi-ma-b0-gf4-a0-89.ucode    TAS2XXX11541.bin
8E86-2-9.bin                       iwlfwifi-ma-b0-gf4-a0.pnvm        TAS2XXX12040.bin
8E86-2-A.bin                       iwlfwifi-ma-b0-gf-a0-89.ucode     TAS2XXX12041.bin
8E86-2-C.bin                       iwlfwifi-ma-b0-gf-a0.pnvm         TAS2XXX12140.bin
8E86-2-D.bin                       iwlfwifi-ma-b0-hr-b0-89.ucode     TAS2XXX12141.bin
8E87-1-9.bin                       iwlfwifi-Qu-b0-hr-b0-59.ucode     TAS2XXX13840.bin
8E87-1-A.bin                       iwlfwifi-Qu-b0-hr-b0-72.ucode     TAS2XXX13841.bin
8E87-1-C.bin                       iwlfwifi-Qu-b0-hr-b0-77.ucode     TAS2XXX13940.bin
8E87-1-D.bin                       iwlfwifi-Qu-b0-jf-b0-59.ucode     TAS2XXX13941.bin
8E87-2-9.bin                       iwlfwifi-Qu-b0-jf-b0-72.ucode     TAS2XXX13A40.bin
8E87-2-A.bin                       iwlfwifi-Qu-b0-jf-b0-77.ucode     TAS2XXX13A41.bin
8E87-2-C.bin                       iwlfwifi-Qu-c0-hr-b0-59.ucode     TAS2XXX1EB30.bin
8E87-2-D.bin                       iwlfwifi-Qu-c0-hr-b0-72.ucode     TAS2XXX1EB31.bin
8EF5-0-9.bin                       iwlfwifi-Qu-c0-hr-b0-77.ucode     TAS2XXX1EB3.bin
8EF5-0-A.bin                       iwlfwifi-Qu-c0-jf-b0-59.ucode     TAS2XXX2234.bin
8EF5-0-C.bin                       iwlfwifi-Qu-c0-jf-b0-72.ucode     TAS2XXX2326.bin
8EF5-0-D.bin                       iwlfwifi-Qu-c0-jf-b0-77.ucode     TAS2XXX3870.bin
8EF6-2-9.bin                       iwlfwifi-QuZ-a0-hr-b0-59.ucode    TAS2XXX387D.bin
8EF6-2-C.bin                       iwlfwifi-QuZ-a0-hr-b0-72.ucode    TAS2XXX387E.bin
advansys                           iwlfwifi-QuZ-a0-hr-b0-77.ucode    TAS2XXX387F.bin
agere_ap_fw.bin                    iwlfwifi-QuZ-a0-jf-b0-59.ucode    TAS2XXX3880.bin
agere_sta_fw.bin                   iwlfwifi-QuZ-a0-jf-b0-72.ucode    TAS2XXX3881.bin
```

Lab 6 evidence: `/lib/firmware` listing

9. Labs 7 and 8

Network enumeration (Lab 7) and the enumeration script (Lab 8) are not included because no screenshots were provided for them.

10. Summary

Mistake	Fix
Wrong variable case (<code>DESKTOP_session</code>)	Use <code>DESKTOP_SESSION</code>
Nonexistent variable (<code>XDG_DESKTOP_TYPE</code>)	Use <code>loginctl</code>
Typos in paths (<code>source.list</code> , <code>kalisources</code> , <code>frimware</code>)	Check names with <code>ls</code> , use Tab completion

Skills practiced: system identification, environment variables, systemd services, listening-port checks, hardware enumeration, reading kernel logs, and troubleshooting typos and path errors.

Security relevance: Knowing the OS version, trusted repositories, running services, and open ports defines a system's attack surface. This is the starting point for both penetration testing and hardening.

Sensitive values (Machine ID, Boot ID, disk UUID) were left out of this report.